

Trainings ▾

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

WARNING

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

WARNING

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

WARNING

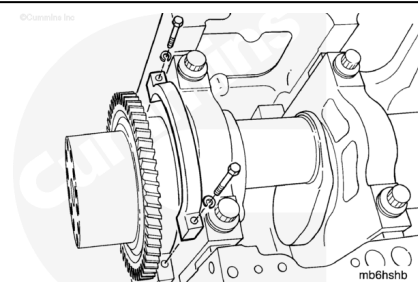
This component or assembly weighs greater than 23 kg [50 lb] or more. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment lift this component or assembly.

- Disconnect the batteries. See equipment manufacturer service information.
- Disconnect the air/fuel supply to the air starter, if equipped. Refer to Procedure 012-022 in Section 12. (</qs3/pubsys2/xml/en/procedures/102/102-012-022.html>)
- Drain the oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/28/28-007-037-tr.html>)
- Remove the full flow filters. Refer to Procedure 007-013 in Section 7. (</qs3/pubsys2/xml/en/procedures/28/28-007-013-tr.html>)
- Remove the oil suction tube. Refer to Procedure 007-035 in Section 7. (</qs3/pubsys2/xml/en/procedures/28/28-007-035-tr.html>)
- Remove the oil pan adapter. Refer to Procedure 007-027 in Section 7. (</qs3/pubsys2/xml/en/procedures/28/28-007-027-tr.html>)
- Remove the engine and place on an engine stand. Refer to Procedure 000-001 in Section 0. (</qs3/pubsys2/xml/en/procedures/28/28-000-001-tr.html>)

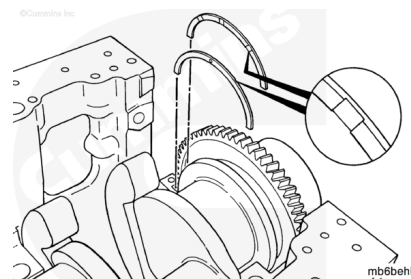
- Remove the flexplate, if installed. Refer to Procedure 016-004 in Section 16.
(/qs3/pubsys2/xml/en/procedures/28/28-016-004-tr.html)
- Remove the flywheel, if installed. Refer to Procedure 016-005 in Section 16.
(/qs3/pubsys2/xml/en/procedures/28/28-016-005-tr.html)
- Remove the flywheel housing. Refer to Procedure 016-006 in Section 16.
(/qs3/pubsys2/xml/en/procedures/28/28-016-006-tr.html)
- Remove the front gear cover. Refer to Procedure 001-031 in Section 1.
(/qs3/pubsys2/xml/en/procedures/28/28-001-031-tr.html)
- Remove the connecting rod bearing caps. Refer to Procedure 001-054 in Section 1.
(/qs3/pubsys2/xml/en/procedures/28/28-001-054-tr.html)
- Remove the main bearing caps. Refer to Procedure 001-006 in Section 1.
(/qs3/pubsys2/xml/en/procedures/28/28-001-006-tr.html)
- Remove the cylinder head. Refer to Procedure 002-004 in Section 2
(/qs3/pubsys2/xml/en/procedures/28/28-002-004-tr.html).
- Remove the piston and connecting rod assembly. Refer to Procedure 001-054 in Section 1.
(/qs3/pubsys2/xml/en/procedures/28/28-001-054-tr.html)

Remove

Remove the two capscrews. Remove the lower thrust bearing support.

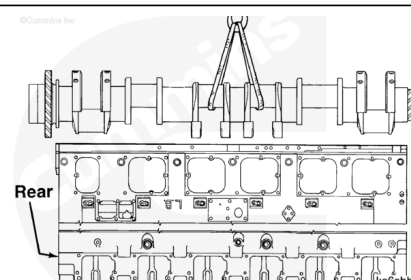


Remove and mark both crankshaft thrust bearings.



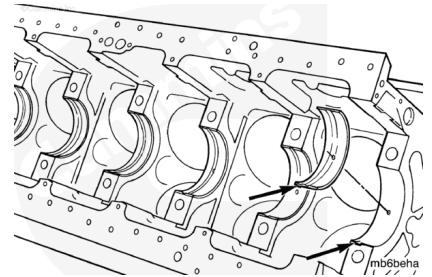
⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb] or more. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment lift this component or assembly.



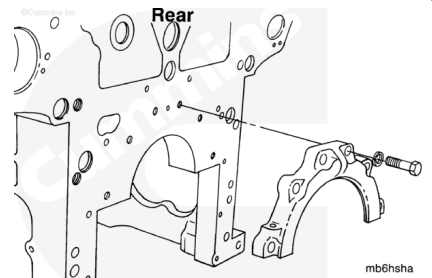
Use a nylon sling and a hoist to remove the crankshaft.

Remove and mark the upper main bearings. Mark the location of the bearing on the tang.



Remove the six capscrews.

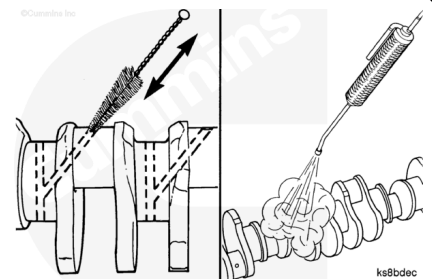
Remove the crankshaft thrust bearing support.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the crankshaft using solvent and dry with compressed air.

Inspect the crankshaft for nicks or scratches. If a mark can **not** be felt with a fingernail, there is no value in polishing the crankshaft.

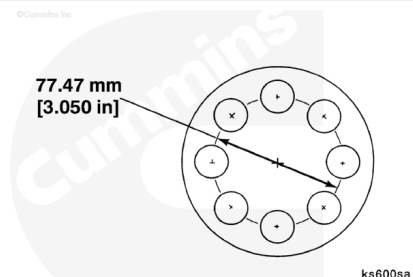
Use a Scotch-Brite™ 7448 abrasive pad, Part Number 3823258, or equivalent, to remove discoloration or light scratches from the machined surfaces.

The crankshaft **must** be cleaned again if this polishing is performed.



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⚠ WARNING ⚠

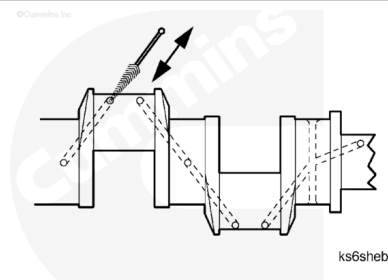
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

The K2000E, K1800E, and K1500E engines use a crankshaft having a 77.47 mm [3.050 in] diameter bolt circle in the nose. The threads are rolled, **not** cut, as on the earlier K50 crankshafts.

To clean the rolled threads, flush with solvent. Dry the threads with compressed air. If additional cleaning is required, brush with a nylon bristle brush.

⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

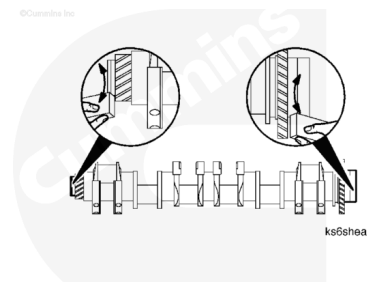


Use solvent and a bristle brush to clean all oil drillings.

Use a light preservative oil to lubricate the part to prevent rust.

⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

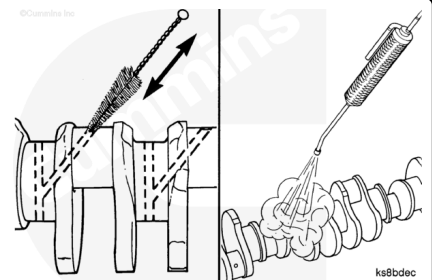


Note : New crankshafts are coated with a heavy preservative. Use solvent to thoroughly remove the coating. Brush or flush the packing debris from the oil drillings before installing the crankshaft in the engine.

Use a hone stone to polish the outside diameter surface at the front and rear oil seal locations, the flywheel mounting location, and the vibration damper location. Remove all small scratches and grooves.

⚠ WARNING ⚠

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⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use solvent to clean the crankshaft.

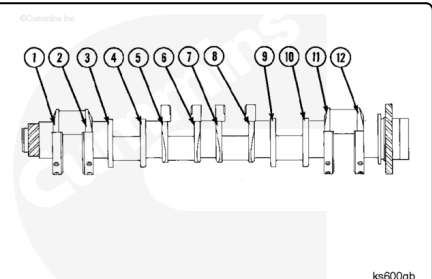
Dry with compressed air.

Note : Make sure to blow out the threaded holes on each end of the crankshaft and the oil drillings. Use a light preservative oil to prevent rust during engine rebuild. Tape over the thread holes.

The webs of the crankshaft are numbered from the front of the crankshaft. The number is steel-stamped on the web.

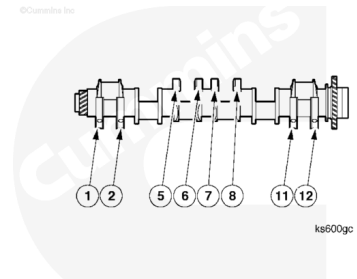
The K50 crankshaft contains 16 webs (not shown).

The K38 crankshaft contains 12 webs, as illustrated.



⚠ CAUTION ⚠

If the counterweights are removed, they must be installed in the correct position on the same crankshaft. If they are installed in the wrong position, engine damage can occur.



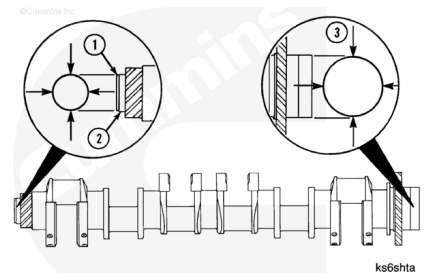
K50 crankshafts contain counterweights that are bolted. The counterweights have the same number as the web to which they are attached. The K50 crankshaft contains eight counterweights.

If a counterweight **must** be replaced, the crankshaft **must** be balanced again. See the Alternative Repair Manual, Bulletin 3379035.

Newer crankshafts contain a ring dowel for the counterweights. The counterweights are **not** interchangeable between the newer and older crankshafts.

Measure the outside diameter at the locations shown.

Crankshaft Outside Diameter			
	mm		in
Location (1)	110.74	MIN	4.360
	110.77	MAX	4.361
Location (2)	111.07	MIN	4.373
	111.13	MAX	4.375
Location (3)	184.10	MIN	7.248
	184.15	MAX	7.250



Check the threads for damage at both ends of the crankshaft.

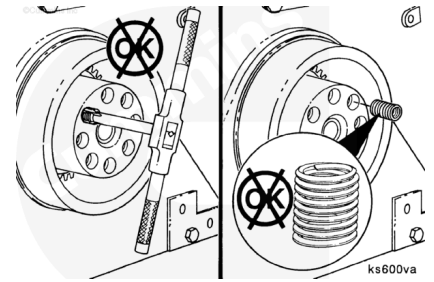
If necessary, see the Alternative Repair Manual, Bulletin 3379035, for repair instructions.

Check the roll pins (4) for damage. Replace the pins if they are damaged.

The roll pins **must** be installed because they align the vibration damper and the flywheel with the crankshaft. The vibration damper and flywheel contain the index marks required for proper adjustment of the valves and injectors.

⚠ CAUTION ⚠

K2000E, K1800E, and K1500E engines use ROLLED threads in the crankshaft nose. Do not use a thread cutting tap to chase or clean the threads, or damage or failure can result.



Thread salvage inserts are **not** permitted in the crankshaft nose.

Inspect the crankshaft seal contact areas for wear. The wear groove is acceptable if it is continuous and no more than 0.12 mm [0.05 in] deep.

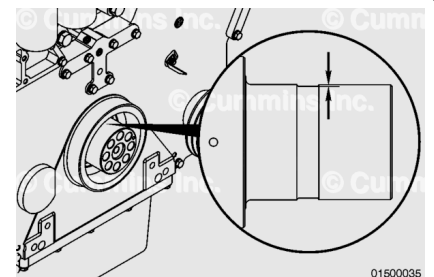
If the wear groove depth is greater than the specification above, the crankshaft **must** be replaced.

If the wear groove depth is less than the specification above, a wear sleeve can be installed to repair the surface. Use the following procedure for front crankshaft seal installation with a wear sleeve. Refer to Procedure 001-023 in Section 1.
(/qs3/pubsys2/xml/en/procedures/28/28-001-023-tr.html)

Installation of a repair sleeve to the front crankshaft nose is recommended during rebuild to protect the crankshaft nose.

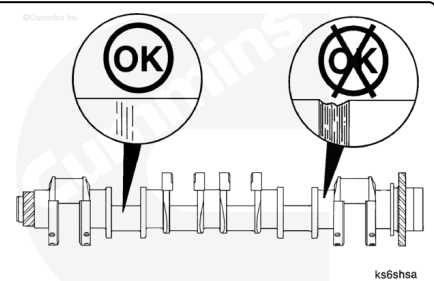
A repair sleeve is **not** available for the rear crankshaft nose as it is already incorporated into the rear crankshaft seal.

Note : The crankshaft nose must not be repaired using any metal spray or welding process.



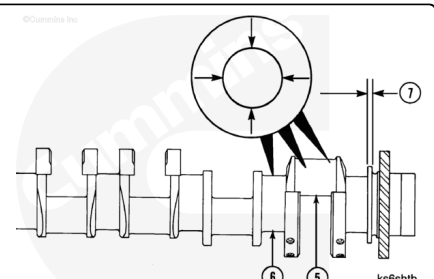
Check the main bearing journals and the rod bearing journals for damage or excessive wear. Minor scratches are acceptable.

If a fingernail catches in the scratch, the crankshaft **must** be replaced or machined. See the Alternative Repair Manual, Bulletin 3379035.



Measure the outside diameter of all the journals and the thrust flange thickness.

Crankshaft Main Bearing Journal Outside Diameter (6)			
	mm		in
Standard	165.05	MIN	6.498
	165.10	MAX	6.500



Crankshaft Main Bearing Journal Outside Diameter (6)			
	mm		in
Undersize 0.25 mm [0.010 in]	164.80	MIN	6.488
	164.85	MAX	6.490
Undersize 0.50 mm [0.020 in]	164.55	MIN	6.478
	164.60	MAX	6.480
Undersize 0.76 mm [0.030 in]	164.29	MIN	6.468
	164.34	MAX	6.470
Undersize 1.016 mm [0.040 in]	164.03	MIN	6.458
	164.08	MAX	6.460

Crankshaft Connecting Rod Journal Outside Diameter (5)			
	mm		in
Standard	107.87	MIN	4.247
	107.95	MAX	4.250
Undersize 0.25 mm [0.010 in]	107.62	MIN	4.237
	107.70	MAX	4.240
Undersize 0.50 mm [0.020 in]	107.37	MIN	4.227
	107.45	MAX	4.230
Undersize 0.75 mm [0.030 in]	107.11	MIN	4.217
	107.19	MAX	4.220
Undersize 1.016 mm [0.040 in]	106.85	MIN	4.207
	106.93	MAX	4.210

Note : The crankshaft can be machined undersize if the outside diameter is not within specifications. Always grind all the journals when one is not within specifications.

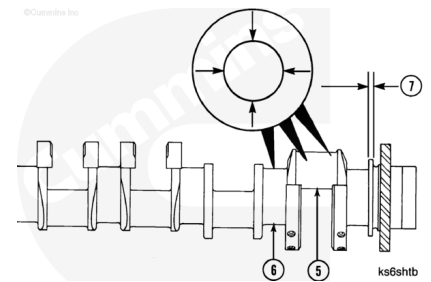
Oversize rod bearings and main bearings are available. See the Alternative Repair Manual, Bulletin 3379035, for grinding specifications and instructions.

Note : Oversize thrust bearings are available if the thrust distance is **not** within specifications. Bearings that are 0.25 mm [0.010 in], 0.51 mm [0.020 in], 0.76 mm [0.030 in], and 1.02 mm [0.040 in] thicker than standard are available. If the crankshaft must be machined for an oversize thrust bearing, see the **Alternative Repair Manual, Bulletin 3379035**.

Use a light preservative oil to lubricate the crankshaft to prevent rust. If the crankshaft is **not** going to be installed immediately, use a heavy preservative oil.

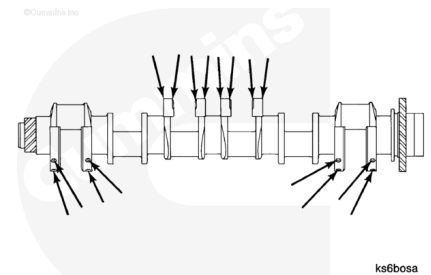
Measure the thrust flange.

Thrust Flange (7)		
mm		in
12.52	MIN	0.493
12.75	MAX	0.502



⚠ CAUTION ⚠

The capscrews must be tight or the counterweight can fall, causing severe damage to the engine. Any loose capscrews must be replaced with new capscrews.



⚠ CAUTION ⚠

Capscrews cannot be used again and must be replaced with new capscrews.

⚠ CAUTION ⚠

Check new capscrews for any visible damage or wear. Discard the capscrew if there is any visible damage or wear to the thread, neck, or capscrew head.

Use a torque wrench to check the torque of the counterweight capscrews. Each counterweight contains two capscrews.

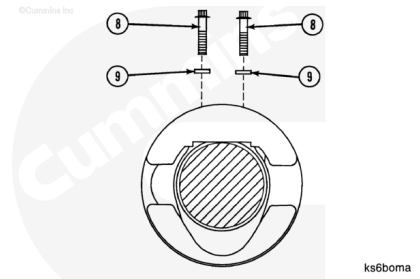
Torque Value: 285 n·m [210 ft-lb]

Mark any counterweights that have a loose capscrew.

Note : The following step applies **only** to the counterweights that contain a loose capscrew.

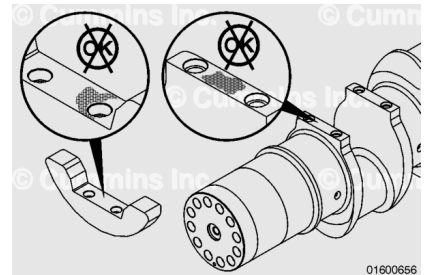
Remove the capscrews (8) and washers (9).

Remove the counterweight.



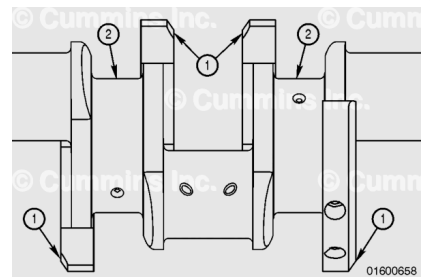
Check the counterweight and crankshaft for fretting or damage between the mating surfaces. If the counterweight is damaged, the counterweight **must** be replaced and the crankshaft **must** be balanced. If the crankshaft is damaged, the crankshaft **must** be replaced or the counterweight replaced and the crankshaft repaired.

Not more than 0.51 mm [0.020 in] can be removed from the crankshaft mounting pad to repair fretting.



⚠ CAUTION ⚠

If the counterweight is removed, it must be installed with the part number toward the nearest main bearing journal. If the counterweight is not installed correctly, the weight can strike the connecting rod, causing extensive damage to the engine.



Note : Make sure the counterweight is returned to its original location. The counterweights have the same number stamped on them as the web to which they are attached.

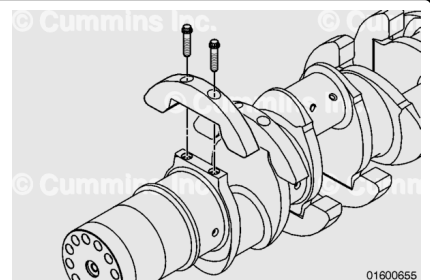
There **must not** be an interference between the crankshaft and counterweight and the sides of the mounting pad.

Lubricate the washers and capscrews with clean engine oil.

Install the washers (9) and capscrews (8).

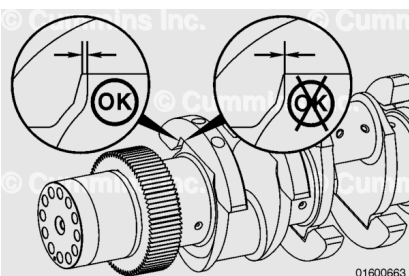
Tighten the capscrews.

Torque Value: 285 n·m [210 ft-lb]



With the capscrews tightened, visually inspect the counterweight for a interference condition with the crank.

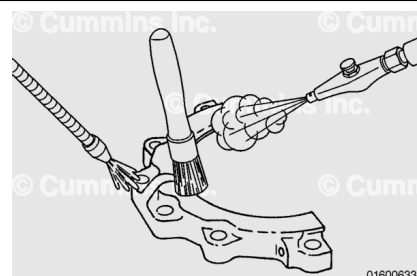
There **must not** be an interference between the crankshaft and counterweight and the sides of the mounting pad.



Inspect the mating surface of the thrust bearing support for damage.

Use emery or crocus cloth, if necessary, to remove any burrs or other imperfections.

Clean the thrust bearing support and retainer thoroughly.



Magnetic Crack Inspect

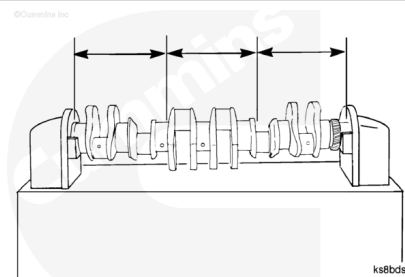
Use a magnetic particle testing machine.

Perform the head shot and inspection method. Then perform the coil shot and inspection method.

Adjust the machine to 1800 ampere VDC or rectified VAC.

Use the continuous method. Wet **only** 1/3 of the crankshaft at a time.

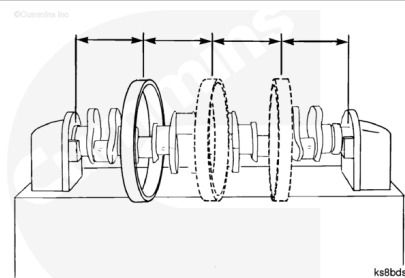
Check the crankshaft for cracks.



Note : Ampere turn is an electrical current of one ampere flowing through the coil, multiplied by the number of turns in the coil.

Use the coil shot method. Use a coil that is a minimum of 514 mm [20.250 in] in diameter.

Use the continuous method to apply coil shot.



Amperage - (Ampere Turns)	
Minimum	4500 VDC or Rectified VAC
Maximum	5000 VDC or Rectified VAC

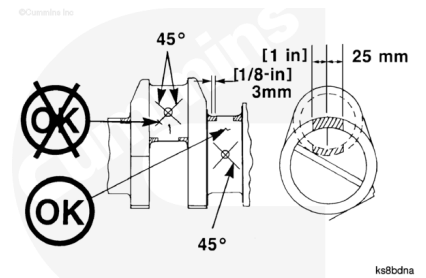
Check the crankshaft for cracks.

Limits of Acceptance - Open Indications:

An open indication is visible to the eye after the wetting operation has been completed. An indication below the surface is **not** visible to the eye after the wetting operation has been completed. An indication below the surface can be seen with the use of the ultraviolet light that is part of the machine.

Do **not** use the crankshaft if:

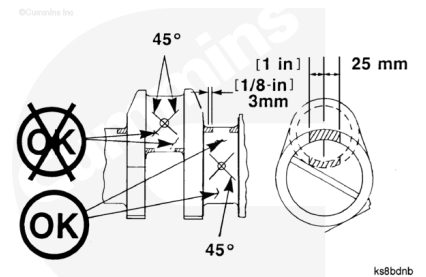
- There is an indication in the fillet or in the shaded area
- There is an indication that passes through the 45 degree diagonal from the oil hole or goes into the oil hole chamfer
- There is an indication that is longer than 6 mm [0.250 inch]
- There are more than four indications on one journal.



Limits of Acceptance - Indications Below the Surface:

Do **not** use the crankshaft if:

- There is an indication in the fillet or in the shaded area that is in a circumferential direction.
- There is an indication in a circumferential direction that is longer than 25.4 mm [1.00 in].
- There is an indication in an axial direction that is longer than 9.5 mm [0.375 in].
- There is an indication that is closer than 1.5 mm [0.0625 in] to an oil hole chamfer.
- There is an indication that passes through the 45 degree diagonal from the oil hole.

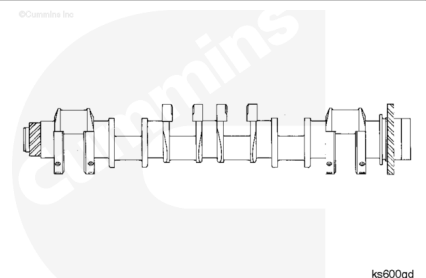


⚠ WARNING ⚠

Wear protective clothing to prevent personal injury from burns.

⚠ CAUTION ⚠

The crankshaft must be demagnetized completely and cleaned thoroughly. Small metal particles can cause engine damage.



Demagnetize the crankshaft.

Use steam to clean the crankshaft and the oil drillings.

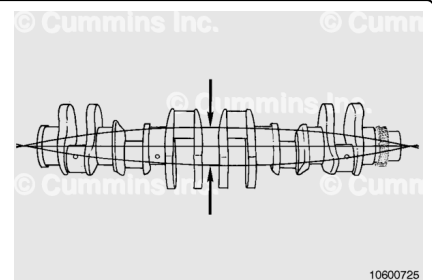
Use a light preservative oil to lubricate the crankshaft to prevent rust.

Note : Use a heavy preservative oil if the crankshaft is not going to be installed immediately. Protect the part with a cover to prevent dirt from sticking to the oil.

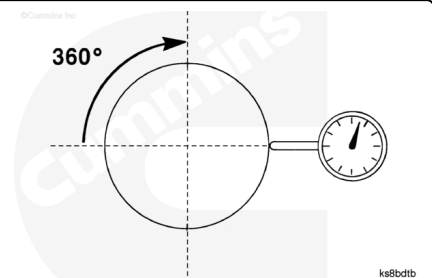
Bend and Twist Inspect

The crankshaft straightness is determined by the amount of total runout and the adjacent journal runout.

The following definitions will help in understanding the procedure:

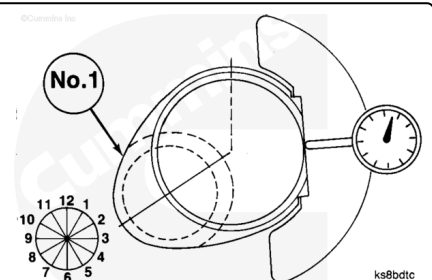


- **Journal Runout:** The top indicator reading (total sweep of the needle) of the main bearing journal as the crankshaft is rotated one complete revolution (360 degrees). Total runout refers to the maximum deflection taken at the journal in the center.
- **Adjacent Journal Runout:** The relationship of the top indicator reading of a main bearing journal as it is rotated on a common axis to the top indicator reading of an adjacent journal.



Adjacent Journal Runout is often referred to as the step runout, the bearing-to-bearing runout, or the journal-to-journal runout.

- **Clock Position:** The location of the journal at the highest top indicator reading point. Compare its angular relationship with the number 1 crankshaft pin as viewed from the front of the crankshaft. In this example, the crankshaft pin is at the 8 o'clock position. This is the clock position of the journal being measured.



Measurement of crankshaft straightness with the specified accuracy can **only** be achieved if the crankshaft:

- Is correctly supported
- Is at thermal equilibrium
- Can be rotated smoothly.

These requirements can **not** be met by measuring the crankshaft while supported by bearing shells in a cylinder block. Rotation of the crankshaft with such an arrangement will lead to errors as the result of lateral and vertical displacement arising from curvature and clearance in the main bearings.

Measurement Environment

- Measurement **must** be conducted in a temperature controlled room at a temperature of 20°C [68°F] +/- 1 degree.
- The crankshaft to be measured **must** be stored in this room for at least twenty four hours to allow the crankshaft to stabilize at room temperature.
- Temperature of the crankshaft **must** be verified before measurements starts, and recorded on the measurement data worksheet included in this document. See measurement data worksheet section 1.8. below.

Measurement Setup

- The measurement table **must** be rated to support at least 450 kg [992 lb] and provide a stable, flat surface that allows V-blocks to be aligned and secured with the crankshaft centerline in a straight horizontal position.

The use of a granite measurement table and roller V-blocks are recommended. See Figure 1. Roller V-blocks have lower rolling resistance than standard V-blocks when rotating the crankshaft during measurement process.

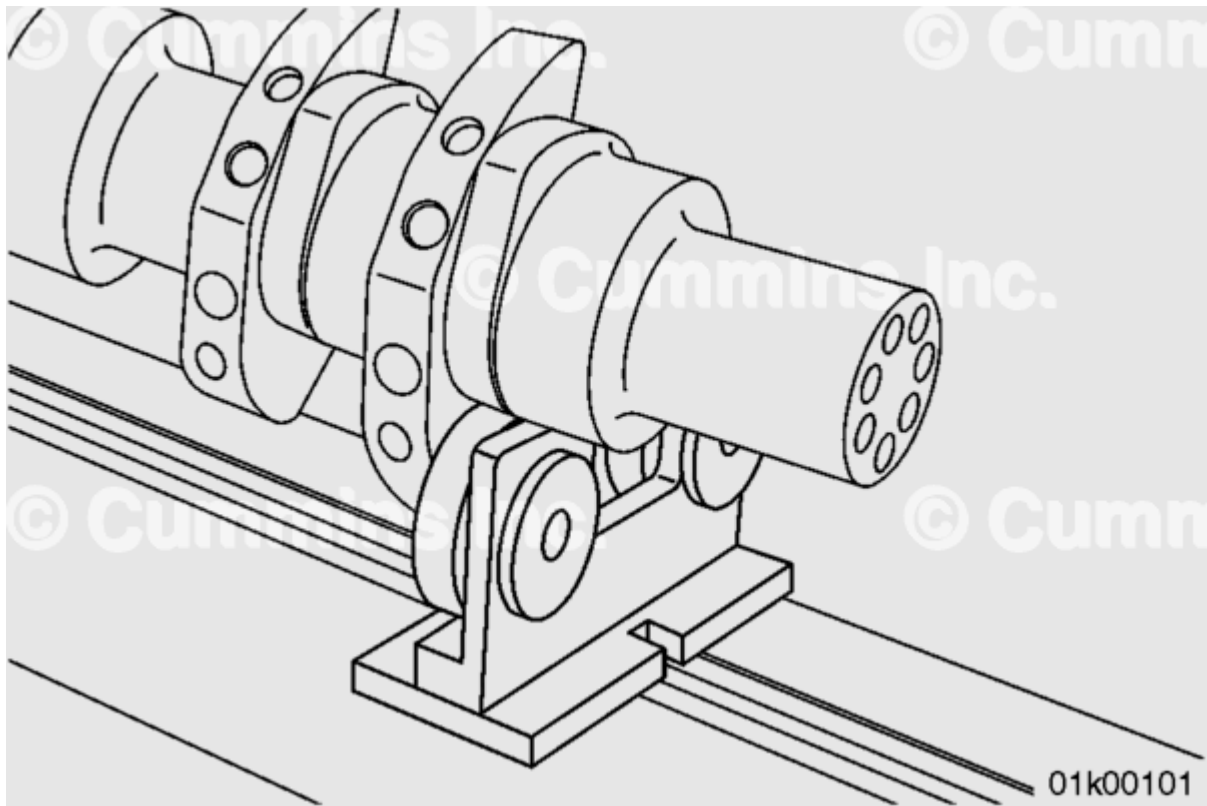


Figure 1.

Figure 1: Crankshaft on Roller V-blocks

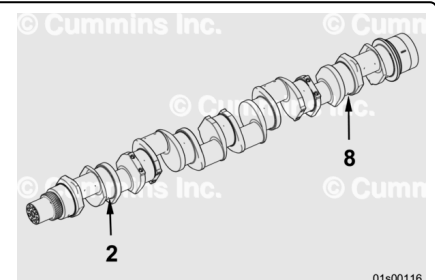
- A motorized drive is preferred to be used to rotate the crankshaft smoothly over the full 360 degrees of rotation. Direct manual rotation **must** be done very carefully to provide the smooth motion necessary for consistent measurements.
- The height of V-blocks **must** be equal (Z-axis). The height difference between the V-blocks **must** be much smaller than part specified tolerance.

Example: If straightness specification is less than 0.06 mm [0.002 in] then the height difference **must** be 0.01 mm [0.0004 in] or less. The V-blocks height **must** be verified in the contact points of V-blocks with the crankshaft journal using a calibrated shaft or bar. Measurements can be made with a mechanical dial indicator or an electronic precision measurement system.

- The measuring device used **must** have a valid calibration certificate and **must** have resolution, repeatability, and accuracy (relationship of the real dimension to what the measurement device indicates) appropriate for the straightness specification being evaluated.

Place two V-blocks on a flat surface. Support the crankshaft on the V-blocks on main journal number 2 and main journal number 8, as shown.

Verify that the crankshaft is free **only** to rotate, with no vertical or lateral motion as the result of clearance in the rollers or rocking of the V-blocks.

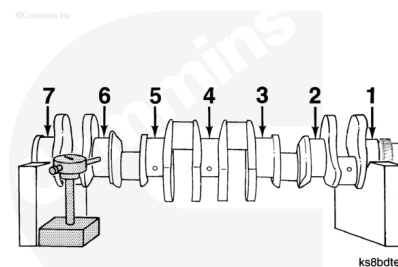


Use a dial indicator to position the indicator so the stem touches the bearing journal being measured at the journal centerline.

The point of measurement chosen **must** be used for measuring all bearing journals.

The K50 crankshaft has nine main bearing journals (not shown).

The K38 crankshaft has seven main bearing journals, as illustrated.



Rotate the crankshaft and measure the top indicator reading at each bearing journal. Record the value and the clock position for each journal.

ITEM	RUNOUT TIR [INCH]	CLOCK POSITION
JOURNAL STEP		
1	[0]	0
2	[0.0021]	12
3	[0.0030]	12
4	[0.0039]	1
5	[0.0025]	1
6	[0.0016]	2
7	[0]	0

Check the part number to determine if the crankshaft is a fully fillet hardened type, or is a non-fully fillet hardened type. Refer to Procedure 001-999 in Section F.

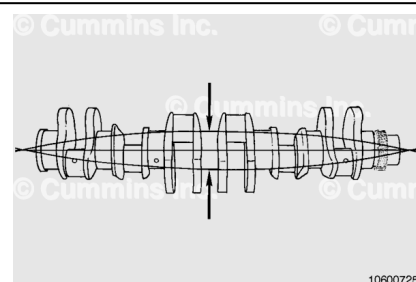
(/qs3/pubsys2/xml/en/procedures/28/28-001-999-tr.html)

A fully fillet hardened or shot-peened crankshaft **must** be discarded if the total runout is **not** within specifications. A crankshaft that has no fillet treatment can be straightened. Refer to the Alternative Repair Manual, Bulletin 3379035.

Determine the Crankshaft Total Runout

Compare the runout to the specifications.

Crankshaft Total Runout				
	mm		in	
K38 Engine		0.203	MAX	0.0081
K50 Engine		0.267	MAX	0.0105

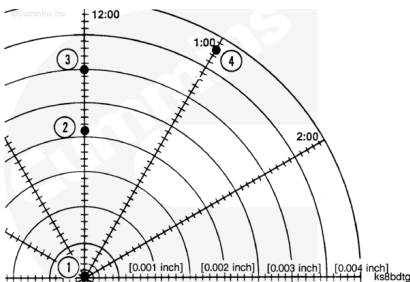


Determine the Adjacent Journal Runout:

The example illustrates the following plot points.

For each journal, plot the top indicator reading value at its clock position on a polar chart illustrated on the following page.

The end journals supported by V-blocks **must** be plotted at the center of the chart.



Journal	Top Indicator Reading	Clock Position
1	0	0
2	0.0021	12
3	0.0030	12
4	0.0039	1

Draw a straight line between the plotted points, (from journal number 1 to number 2, and from number 2 to number 3), until all journals are plotted on the chart.

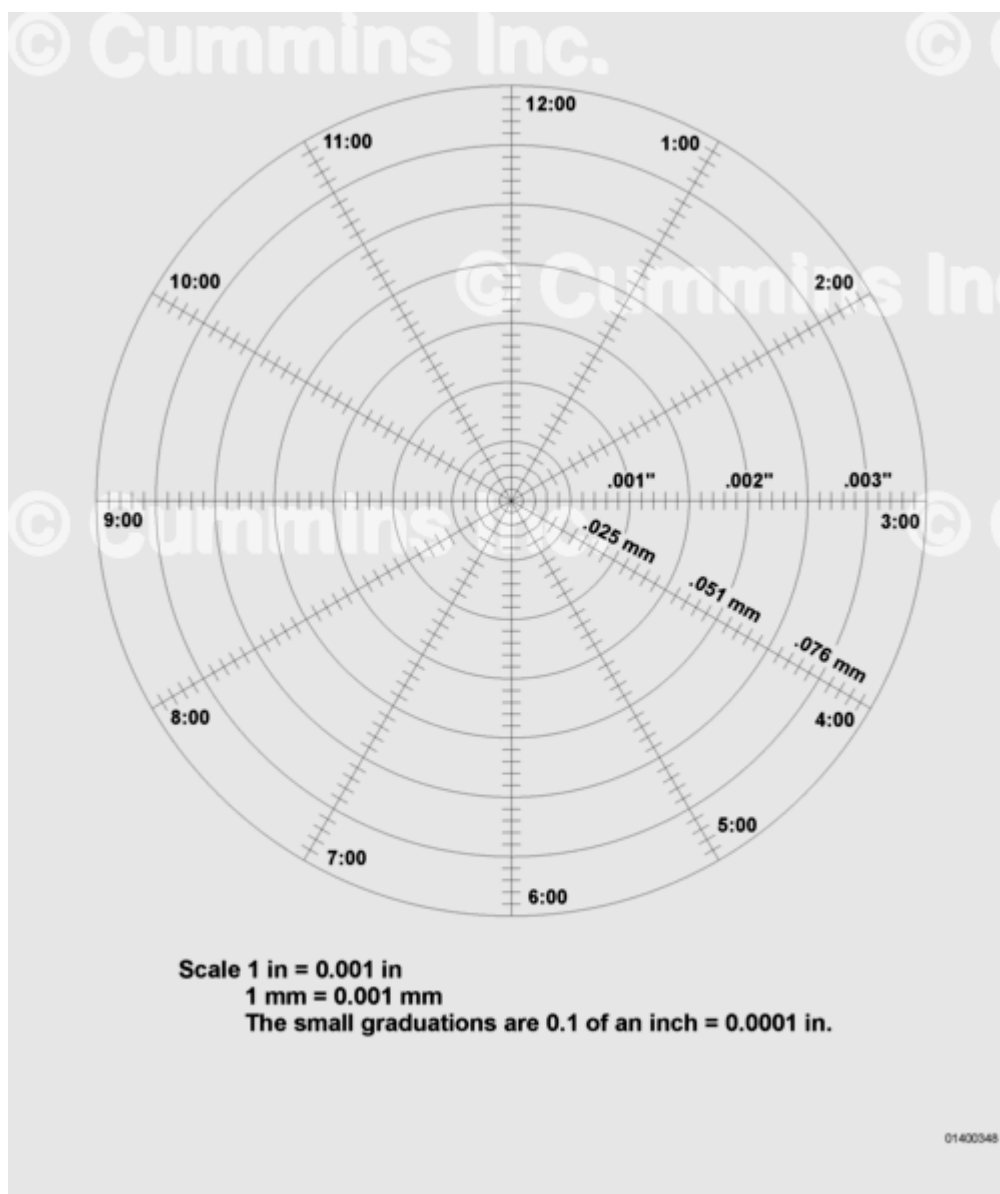
To determine the adjacent journal runout, measure the length of the line from each journal to its corresponding journal point.

Illustrated as an example, journal number 3 to number 4 is 5.1 mm [2 in]. This represents a runout of 0.051 mm [0.002 in].

Record the adjacent journal runout for each bearing journal. Compare the adjacent journal runout to specifications.

Adjacent Journal Runout				
	mm		in	
Fully Fillet Hardened		0.089	MAX	0.0035
Non-Fully Fillet Hardened		0.089	MAX	0.0035

Replace the crankshaft if any measurement is **not** within specifications.



Polar Chart

Scale 1 inch = 0.001 inch and 1 mm = 0.001 mm

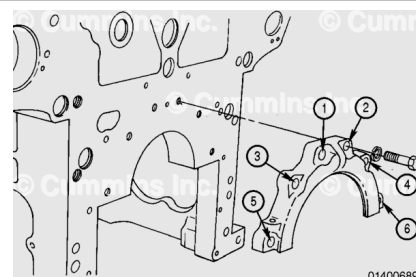
The small graduations are 0.1 of an inch = 0.0001 in.

Install

Install the crankshaft thrust bearing support to the cylinder block.

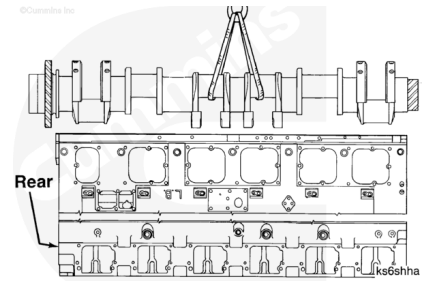
Tighten the six capscrews and lock washers in the sequence shown in the illustration.

Torque Value: 60 n•m [44 ft-lb]



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment lift this component or assembly.



Note : Complete the upper main bearing installation prior to installing the crankshaft. Refer to Procedure 001-006 in Section 1.
(/qs3/pubsys2/xml/en/procedures/28/28-001-006-tr.html)

Use a lint-free cloth. Clean the crankshaft bearing journals.

Note : The end of the crankshaft with the gear that has the smallest diameter must point toward the front of the block.

Install the crankshaft.

Finishing Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment lift this component or assembly.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the thrust bearings. Refer to Procedure 001-007 in Section 1.
(/qs3/pubsys2/xml/en/procedures/28/28-001-007-tr.html)
- Install the main bearing caps. Refer to Procedure 001-006 in Section 1.
(/qs3/pubsys2/xml/en/procedures/28/28-001-006-tr.html)
- Install the cylinder head. Refer to Procedure 002-004 in Section 2
(/qs3/pubsys2/xml/en/procedures/28/28-002-004-tr.html).
- Install the piston and connecting rod assembly. Refer to Procedure 001-054 in Section 1.
(/qs3/pubsys2/xml/en/procedures/28/28-001-054-tr.html)

- Install the connecting rods. Refer to Procedure 001-054 in Section 1.
(/qs3/pubsys2/xml/en/procedures/28/28-001-054-tr.html)
- Install the front gear cover. Refer to Procedure 001-031 in Section 1.
(/qs3/pubsys2/xml/en/procedures/28/28-001-031-tr.html)
- Install the flywheel housing. Refer to Procedure 016-006 in Section 16.
(/qs3/pubsys2/xml/en/procedures/28/28-016-006-tr.html)
- Install the flywheel, if removed. Refer to Procedure 016-005 in Section 16.
(/qs3/pubsys2/xml/en/procedures/28/28-016-005-tr.html)
- Install the flexplate, if removed. Refer to Procedure 016-004 in Section 16.
(/qs3/pubsys2/xml/en/procedures/28/28-016-004-tr.html)
- Remove the engine from the engine stand and install. Refer to Procedure 000-002 in Section 0.
(/qs3/pubsys2/xml/en/procedures/28/28-000-002-tr.html)
- Remove the thrust bearings. Refer to Procedure 001-007 in Section 1.
(/qs3/pubsys2/xml/en/procedures/28/28-001-007-tr.html)
- Install the oil pan adapter. Refer to Procedure 007-027 in Section 7.
(/qs3/pubsys2/xml/en/procedures/28/28-007-027-tr.html)
- Install the oil suction tube. Refer to Procedure 007-035 in Section 7.
(/qs3/pubsys2/xml/en/procedures/28/28-007-035-tr.html)
- Install new full flow filters. Refer to Procedure 007-013 in Section 7.
(/qs3/pubsys2/xml/en/procedures/28/28-007-013-tr.html)
- Fill the engine with clean 15W-40 lubricating oil. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/28/28-007-037-tr.html)
- Connect the air/gas supply to the air starter, if equipped. Refer to Procedure 012-022 in Section 12. (/qs3/pubsys2/xml/en/procedures/102/102-012-022.html)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine. Check for leaks.